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FPGA-Based Design and Implementation of Cascaded Digital Filters for Enhanced Auscultation in Digital Stethoscopes

S. Devi Poonguzhali[1] · **T. N. Prabakar**[1] · **V. Elamaran**[1]

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Abstract
This paper describes the design, implementation and systematic validation of a new field programmable gate array (FPGA) based cascaded digital filter architecture designed to considerably improve auscultation in digital stethoscopes through the efficient rejection of noise and amplification of vital body sounds. The proposed cascaded filter unit consisting of three discrete-time filters is designed in such a way that one to remove motion noise, the second to isolate circulatory and respiratory sound while rejecting the background noise and the third for isolating heart sounds. The filters that are suggested were thoroughly designed and tested on two platforms: MATLAB as a starting point for simulation and validation and a FPGA both for software modeling and hardware implementation. On the FPGA, the stethoscope filters were implemented with two infinite impulse response filter design structures as conventional and direct form I those were evaluated under both non-pipelined and highly efficient pipelined designs. One of the primary contributions of the proposed work is the demonstrable improvement in working speed and throughput processing attained by the pipelined implementation, which greatly minimizes latency over traditional non-pipelined designs. From the hardware verification, with very high speed integrated circuit hardware description language (VHDL) coding on Altera/Intel FPGA 2C20F484C7 board, we carried out a thorough investigation of resource usage, performance (speed), and power consumption. Our findings show remarkable reductions in logic element usage and strong real-time functionality, making this design a very practical and resource-conscious solution for handheld medical equipment. This study not only pushes the state-of-the-art in digital stethoscope technology but also provides a flexible and high-performance platform for real-time adaptive filtering in other wider biomedical applications.

Keywords Digital stethoscope · Auscultation · Digital filters · Pipelining · FPGA · Noise reduction

Extended author information available on the last page of the article

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1 Introduction

According to the American Heart Association, cardiovascular disease is the world's largest reason for death, resulting in around 17.9 million deaths annually. A crucial tool for evaluating heart health is the stethoscope. Auscultation, the listening to internal body sounds, is still a core method of medical diagnosis and its symbol being everywhere the stethoscope. Although acoustic stethoscopes provide the direct, non-invasive method of examining vital physiological functions, their own innate shortcomings such as susceptibility to surrounding noise, limited sound quality and inaptitude for digitally recording or analyzing results frequently hinder precise diagnosis, especially of subtle or high-frequency defects. Moreover, classical acoustic stethoscopes suffer some limitations in sound quality and recording. Their core design relies on direct sound transfer through tubing that can produce inferior sound quality, particularly in crowded areas or during the assessment of high-frequency, quiet sound. Their incapacity to capture, store or disseminate the auscultated sounds further reduces their utility for remote diagnosis, training or retrospective analysis.

Digital stethoscopes address the shortcomings of acoustic stethoscopes by converting acoustic signals into digital data, enabling advanced signal processing for enhanced sound quality and diagnostic utility. The field of auscultation may soon experience a revolution through the creation of advanced digital processing algorithms for digital stethoscopes that will enable healthcare providers to diagnose and assess patients with enhanced accuracy, ultimately improving a patient's outcomes. In our work the digital stethoscopes equipped with advanced processing capabilities, like the cascaded filter unit developed can enhance cardiology practice. Better patient outcomes, better care and possibly cheaper long-term costs can result from these advancements in the detection and diagnosis of cardiovascular illnesses. A phonocardiogram (PCG) signal is a graphic depiction of heart sounds is commonly obtained using a microphone or electronic stethoscope. The PCG monitors the heart's mechanical activity, including murmurs and sounds like S1 and S2, to diagnose valvular diseases. Noisy PCG data, which are used to examine heart sounds, may impair diagnosis accuracy. The primary issue with such a system is the interference that arises when the PCG signal is being acquired (Fig. 1) [9, 20, 24, 26]. These interferences might be of internal origin primarily from respiratory and digestive sounds or external, originating from the environment, such as speech or exterior sounds those brought on by the movements of patients and externally from the environment such as speech or outside noises [24, 26]. In addition to those brought on by patient movements [24, 26], other disruptions can occasionally be caused by coughing, muscular movements, swallowing, sensor rubbing, etc., [20]. The accuracy and dependability of PCG recordings may be impacted by these undesired signals which could make it difficult for medical practitioners to interpret the data and could result in delayed or inaccurate diagnosis. Development of suitable denoising methods for PCG signals improves the heart sound quality in diagnostic systems. [9]

Digital stethoscopes utilize advancements in sensor technology and digital signal processing. They capture sounds from the body using a microphone, converting these sound waves into digital data to enhance sound quality, minimize signal noise and enable digital recording. Digital stethoscopes have the potential to transform auscul-

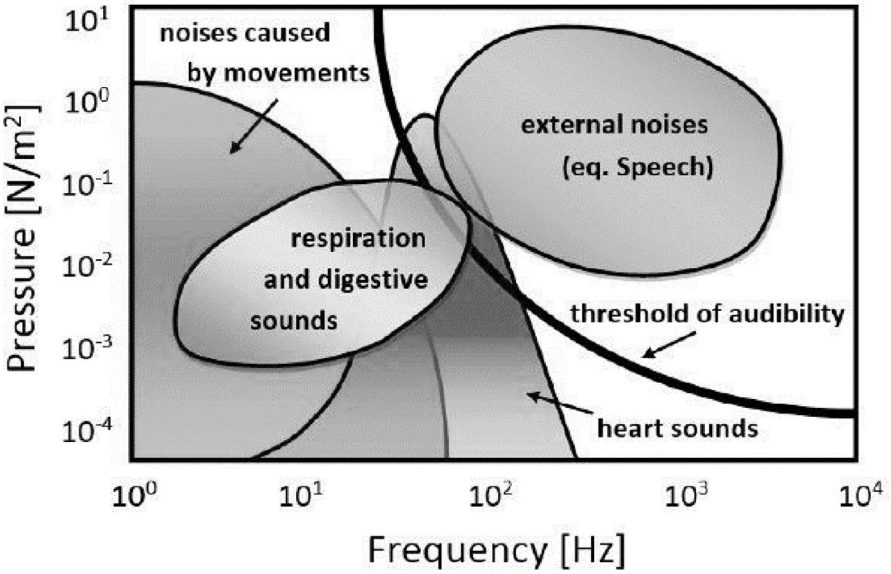


Fig. 1 Contamination of PCG signal with various types of noise (source [9, 24])

tation by providing doctors with more accurate and reliable diagnostic information. A key component of these devices is a digital filter that amplifies the desired sounds while reducing or eliminating unwanted noises and frequencies. However, current digital stethoscope technology often experiences compromises among processing simplicity, real-time capability and resource efficiency from hardware. Most draw upon general-purpose digital signal processors (DSPs) or filtering based on software, which can cause latency, reduce portability due to power requirements or be more expensive for dedicated hardware. In addition, while digital filtering is known to be important for reducing noise and extracting accurate heart sounds, cascading different filters into one unit, remains a major challenge in terms of providing high throughput with low hardware overhead.

This paper introduces a novel FPGA-based cascaded digital filter structure that breaks through current limitations in digital stethoscope auscultation. Taking advantages of the inherent parallelism and re-configurability of FPGAs, our design achieves a three stage digital filtering system with excellent noise reduction, true heart sound extraction, and very minimal hardware resources. The main contributions of this paper are:

1. A new cascaded filtering architecture optimized for FPGA operation with a carefully designed pipeline of digital filter stages that effectively cancel out noise profiles and amplify dominant heart sound frequencies.
2. Outstanding real-time performance via pipelining, alleviating processing delay and making real-time auscultation feasible for use in clinics.

Our VHDL implementation and optimization methods, such as efficient pipelining, translate to considerable logic element making it highly integrable into small, power-limited digital stethoscope devices. Comprehensive MATLAB simulation and FPGA hardware tests establish our cascaded digital filter to be highly capable of rejecting noise and improving fidelity of the physiological signal. This work presents a reliable solution to next-generation digital stethoscopes and lays foundation for high-performance real-time signal processing in handheld medical diagnostic equipment.

The rest of the paper is structured in the following way: Sect. 2 provides an overview of previous works. Materials and Methods are discussed in Sect. 3. Hardware implementation and simulation results are presented in Sects. 4 and 5, respectively. Section 6 gives the results and discussion. The concluding statement of the work is given in Sect. 7.

2 Related Works

Electronic stethoscopes increasingly used for cardiopulmonary auscultation often generate a noise resembling a crackle termed "fake crackle", which interferes with diagnosis. Fake crackles predominantly occur in lower lung regions and during the inspiratory phase, within a frequency range of 250–1950 Hz similar to fine crackles, making them difficult to differentiate by ear. The noise occurs when the stethoscope diaphragm is slightly displaced from the skin, especially during lung expansion, and overlaps with real crackles, reducing diagnostic reliability, highlighting the need for improved electronic stethoscope technology [14].

This paper [2] describes a new approach to reduce baseline wandering noise from electrocardiogram (ECG) signals using fast infinite impulse response (IIR) filters implemented on FPGA devices. Baseline wandering noise can greatly distort ECG signals and thus may distort ECG signals may compromise the interpretation and strength of resulting diagnoses. The authors of this work offer efficient hardware implementations of Direct Current (DC) bias removal filter and also explored the performance and average resource usage. With a speed of 528.54 MHz, the pipelined DC bias reduction filter with look-ahead pipelining technique exhibited the best performance gives the best speed versus resource utilization tradeoff.

Phonocardiogram signals collected using electronic stethoscopes can be compromised by external artifacts such as noise, or body motion. These artifacts can produce ambiguous, or incorrect heart signal interpretation by human and automated systems alike. A technique has been developed [4] which utilizes the tuner-Q-wavelet transform (TQWT) and signal second difference with median filter (SDMF) to detect and remove artifact-contaminated sub-sequences from PCG signals. The goal of the authors is to allow for automation of clean PCG signals-to-cardiac evaluations in a clinical or ambulatory context.

This study [18] assesses the performance of digital filters as part of electronic stethoscopes in order to detect specific heart sound patterns related to diseases and improve diagnostic capabilities. The system includes an analog microphone captures the sound of the heart, which can be amplified, filtered (high-pass and low-pass), processed digitally (IIR) by the Arduino Mega 2560 microcontroller. Eight

healthy subjects were employed to examine the viability of the digital and analog filters. The outcomes showed that digital filter outperformed analog filter regarding signal to noise ratio (SNR). The output from the IIR digital filters had an SNR of 20.43 ± 1.78 dB and the 4th order analog filter had an SNR of 12.5 ± 2.5 dB. Therefore, while both digital and analog filters lowered noise and improved overall gain of the heart sounds, the digital filter substantially reduced noise and delivered considerably clearer heart sound signals for diagnostic consideration.

ECG signal is a significant diagnostic tool for diagnosis of heart diseases through the use of P, QRS and T waveform analysis. R-peak detection is significant in computing heart rate variability (HRV), but movement artifacts or noise may lower accuracy. There are numerous sources of noise such as baseline drifts, power line interference, and electromyography that contaminate ECG signals. Though conventional filters such as finite impulse response (FIR) and infinite impulse response (IIR) filters are good in reducing noise, IIR filters remain in favor because of their ease of computation even if they are noisier. In order to overcome this limitation, the authors in [7] introduced a sophisticated filtering algorithm, the optimum reduced-order IIR filter (ROIIR), which enhances peak-to-peak swing detection by 8.8% from previous designs.

The paper [25] presents a low-cost electronic stethoscope that captures heart sounds as electrical signals to visualize and analyze. The device is equipped with a preamplifier, high-pass and low-pass filters, a microcontroller, and Bluetooth. The heart sounds are amplified 10–30 times and filtered by a fourth-order Butterworth filter that effectively removes noise while preserving the integrity of the signal. The filtered signals are sent to a computer interface (LabVIEW) through Bluetooth. Though the system is able to accurately identify primary heart sounds (S1 and S2), it fails to accurately identify lower-frequency sounds (S3 and S4) and this points towards further optimization in hardware and filter methods.

The study [3] proposes the design of an inexpensive electronic stethoscope for the diagnosis of heart defects through digitized heartbeat sound analysis to offer a low-cost, portable diagnostic tool with accuracy. The device is made up of a condenser microphone, preamplifier and filter circuits and utilizes fourth-order Butterworth LPF and HPF filters due to their sharpness of frequency response and stability. The frequency range of maximum heart sound amplitude is 20–120 Hz. Butterworth filter outperformed Chebyshev and Bessel filters and yielded smooth and ripple-free heart signal patterns and thus were appropriate for this purpose.

The research [16] introduces an electronic-to-acoustic stethoscope filter that aims to modulate the frequency response of electronic stethoscope recordings to simulate the sounds detected with acoustic stethoscopes. Through comparison and analysis of lung sounds obtained with both electronic and acoustic stethoscopes, researchers created this filter. Findings showed that clinicians identified filtered electronic and acoustic stethoscope sound transitions less than unfiltered electronic and acoustic stethoscope sound transitions, which indicates success of the filter. This approach to filtering may ease clinicians' transition to electronic stethoscopes and allow them to realize the benefits of technology without having to abandon familiar sound quality with acoustic stethoscopes.

The research [13] had the objective of designing an optimal infinite impulse response (IIR) filter for classifying ECG signals in detecting heart disorders by eliminating baseline noise and power line interference. The method applied reduced-order IIR filter design using optimization techniques like the min–max approach, where a two-stage filter consisting of passband and stopband filters was used for signal improvement. Combining QRS peak classification and detection in MATLAB offered improved accuracy with an accuracy rate of 96.87%, enhanced efficiency in QRS peak detection, and improved HRV analysis. The analysis, conducted using sophisticated filtering techniques on 30 datasets of ECG, calculated time-domain parameters such as standard deviation of NN intervals (SDNN) and root mean square of successive differences (RMSSD) to analyze HRV. The approach improves cardiac rhythm analysis and early abnormality detection, ensuring effective cardiac health monitoring.

High-speed IIR filter synthesis on FPGA in [22] introduced three new architectures, namely a lossy integrator-based look-ahead IIR filter, a two-level parallel-pipeline IIR filter, and an FIR-based IIR filter. The FIR-based IIR filter proposed on the basis of parallel-pipeline FIR architecture exhibited better hardware utilization, power reduction, and acceleration than traditional IIR filters. At implementation on a Xilinx Virtex-5 FPGA, the FIR-based IIR filter reached its highest frequency of 285.105 MHz and consumed less power than other methods. The method applies fine-grain pipelining, parallel computation, and scaling factors, therefore, it is both low-power consumption and high-speed and can be implemented in real-time signal processing systems.

In [6] SLND (Supervised Learning-Based Noise Detection), to improve the specificity and sensitivity of ECG signal analysis by effectively detecting and handling noise in distributed medical Internet of Things (IoT) environments is proposed. The SLND method analyzes temporal and spectral characteristics of ECG signals, including energy, zero-crossing rates, and Fourier transformation components, to identify noise. An AdaBoost classifier is utilized to distinguish noisy from clean ECG signals, with a statistical test (Kolmogorov–Smirnov test) comparing feature distributions to ensure the best noise detection. The approach performed better than a state-of-the-art model (EMD+NLM) in a comparative study on a benchmark database (MIT-BIH) with more accurate results (precision: 89%, sensitivity: 95%, specificity: 94%) and evidenced its efficacy in noise detection for ECG signals, which could be extended to enhancing the dependability of automated ECG analysis in clinical and IoT based medical monitoring.

A research study [10] investigated the use of discrete wavelet transform (DWT) for denoising ambulatory ECG signals to enhance the quality of the signals for efficient diagnosis of the heart. A DWT-based noise reduction technique employed various wavelet functions like Daubechies, Symlets, and Biorthogonal wavelets with hard and soft thresholding at various levels of decomposition. The performance measures such as Signal-to-Noise Ratio (SNR), Mean Square Error (MSE), and Peak Signal-to-Noise Ratio (PSNR) were utilized to measure the result. It came to be understood that the optimal result was achieved using Daubechies wavelet (Db8) at level 9 with maximum SNR (74%), minimum MSE (0.96), and optimum PSNR through soft thresholding, which enabled identification of heart features such as R-R intervals,

QRS complexes, and heart rate. Db8 and Sym4 wavelet functions yielded best results in de-noising ECG signals and suggested that DWT must be combined with deep learning models such as Convolutional Neural network (CNN) for further improvement in ECG analysis and classification.

A trial used LabVIEW and MATLAB [11] to examine heart sound signals recorded via an electronic stethoscope to circumvent limitations of conventional stethoscopes like environmental interference and subjectivity. Signal processing methods like Fast Fourier transform (FFT) and Wavelet Transform were utilized by the research for noise reduction and feature extraction and computerized waveforms of heart sounds for examining normal and pathological sounds without invasive techniques. The system computed Beats per minute (BPM) to examine heart rate and identify normal and abnormal rhythms. MATLAB was discovered to provide analytical depth, while LabVIEW was preferred to implement real-time devices and visualize. The research established that the tools are efficient for processing heart sound signals, providing promise for self-diagnosis and enhancement in clinical care with computerized simulations.

Multi-channel auscultation greatly enhances respiratory sound classification accuracy over single-channel approaches. The research [17] employs a hybrid convolutional neural networks and long short-term memory (CNN–LSTM) network with Mel-frequency cepstral coefficients (MFCC) features to preserve spatial and temporal sound patterns. The most optimal performance was obtained using four-channel recordings, at an accuracy of 92% and F1-score of 93%. Data augmentation and real patient recordings also improved model robustness in noisy clinical settings. The method holds promising clinical value for the diagnosis of respiratory diseases with higher reliability.

A new sliding window least mean squares (LMS) adaptive filter is introduced [12] to denoise PCG signals. It employs a sliding time-domain window and self-tunes step size on the basis of correlation coefficients. The technique considerably improves mean squared error (MSE), SNR, peak signal-to-noise ratio (PSNR) and correlation as compared to standard LMS filters. It removes stationary and nonstationary noise effectively from normal as well as abnormal PCG signals. Experimental results in real-world scenarios exhibit high accuracy and robustness and can be implemented in embedded medical platforms.

The paper [8] proposes a novel cascaded adaptive noise cancellation (ANC) system optimized by the greater cane rat algorithm (GCRA) to enhance PCG signal quality. It successfully removes uniform, Gaussian, and pink noises from PCG signals using an IIR three-stage filter. GCRA performs better than other metaheuristics such as GOA and DMOA for noise elimination, signal integrity, and computational complexity. Actual PCG data from PhysioNet confirms its better performance on a variety of metrics such as SNR, MSE, and correlation coefficient. The technique also enhances deep learning-based normal and abnormal heart sound classification, demonstrating its clinical value.

A new ear-contactless stethoscope [21] was proposed with a Bluetooth speaker for hygienic cardiac sound monitoring. Deep learning algorithms, particularly a CNN-based Inception Net, were fine-tuned to diagnose valvular heart diseases with accuracy of up to 99.65%. Acoustic, time, and frequency domain features from PCG

signals are utilized by the system for accurate diagnosis. Hardware components are Raspberry Pi, filters, and sensors for real-time data acquisition and visualization. This inexpensive AI-based device holds potential for rural healthcare and pandemic-safe cardiac screening.

This research [19] created a lightweight electronic esophageal stethoscope system for capturing heart and lung sounds intraoperatively. It employed a PDMS (Polydimethylsiloxane)-filled microphone case and adaptive noise reduction to minimize operating room noise. Experimental results indicated an improvement in signal-to-noise ratio from 1.0992 to 1.9745 dB. The system provided real-time waveforms and data storage capabilities for future examination. In general, it presents a noise-immunity and reliable intraoperative monitoring solution.

The electronic stethoscope [15] provided 86.9% accuracy of heart sounds and 73.3% for lung sounds. Noise reduction was carried out with the help of cork insulation and double microphones. Signal processing consisted of FFT analysis and feature extraction by MFCC and principal component analysis (PCA). Classification was carried out by applying ensemble learning methods such as AdaBoost and Bagging. Optimal segmentation utilized 2 s frames (heart) and 5 s frames (lung) with overlapping windows.

3 Materials and Methods

To improve the quality of sound signals recorded by stethoscopes, filters play an essential role. By using the right stethoscope, medical professionals can enhance patient outcomes and ensure accurate diagnoses. Electronic stethoscopes use digital filters, which are usually of one of three broad types: adaptive filters, finite impulse response (FIR) filters, and infinite impulse response (IIR) filters. FIR filters are often preferred because they are inherently stable and can produce a linear phase response. This characteristic is vital in medicine, as phase distortion can compromise diagnostic accuracy. While IIR filters have the potential for steeper roll-off and better noise rejection, they come with challenges related to phase distortion and stability. Adaptive filters offer a real-time solution by adjusting their coefficients dynamically to achieve optimal signal enhancement and noise cancellation. This approach is particularly effective in situations where noise characteristics change over time.

When designing these filters, several key considerations must be handled carefully. First, the cutoff frequency needs to be established, as this determines which range of sound wave frequencies the filter will maintain or attenuate. The complexity of the filter structure, as well as its ability to reject unwanted noise, is directly related to the filter's order. Higher-order filters are more effective at rejecting noise but require more computational power. Lastly, the filter's phase response how much it shifts or delays sound signals is crucial, especially in medical applications. To prevent misinterpretations caused by phase-related artifacts, a linear phase response is often necessary, which poses a significant requirement.

3.1 Problem Statement and Main Contribution

Doctors can hear sounds from the heart and lungs, as well as from the circulatory and respiratory systems, with the aid of a stethoscope, a simple diagnostic instrument. A microphone and analog-to-digital converter are employed in digital stethoscopes to capture signals, while conventional stethoscopes employ direct vibration transmission by a tube. In this study, a cascaded filter unit with three discrete-time filters for a digital stethoscope that is optimized for an 8 kHz sampling rate is designed and implemented using VHDL programming.

3.1.1 Filter-1 to Remove the Patient's Physical Movement

Purpose: The primary function of the first filter is to eliminate low-frequency motion artifacts and baseline wander that can corrupt the stethoscope signal. These disturbances typically arise from patient movement, cable friction, or even the subtle handling of the stethoscope itself during auscultation.

Design Rationale: Motion noise typically appears as very low-frequency components, sometimes below heart and lung sound diagnostic frequency range. Hence, this filter is normally designed in the form of a DC notch filter. Its cut-off frequency is optimized so as to eliminate these spurious low-frequency components without suppressing the important heart and lung sound information contained in higher frequency ranges.

Operation: By permitting frequencies over a certain point to pass through but attenuating those beneath it, the first filter removes gross, motion-induced interference from the signal, yielding a more consistent and workable baseline for following processing. This process is important in avoiding saturation or distortion within subsequent filter stages and in offering a clearer signal.

The stethoscope's or patients physical movements produce noise at very low frequencies (around 0 Hz). This is accomplished by removing movement-related noise using a first order IIR filter that has one pole and one zero. The locations of the poles and zeros on the pole-zero plot for this filter are shown in Fig. 2.

Removal of noise at very low frequencies having a central frequency of 0 rad/sample is possible with the DC notch IIR filter, which has poles at Z = 0.9 and zero Z = 1. This might also be thought of as a narrowband passband high pass filter.

The desired transfer function for this filter becomes,

$$\mathrm{H_1\,(z)} = \frac{\mathrm{y_1\,(z)}}{\mathrm{x\,(z)}} = \frac{1 - \mathrm{z}^{-1}}{1 - \frac{15}{16}\mathrm{z}^{-1}} \tag{1}$$

and the desired time-domain output is expressed as,

$$\mathrm{y\,(n)} = \mathrm{x\,(n)} - \mathrm{x\,(n-1)}\,\frac{15}{16}\mathrm{y\,(n-1)} \tag{2}$$

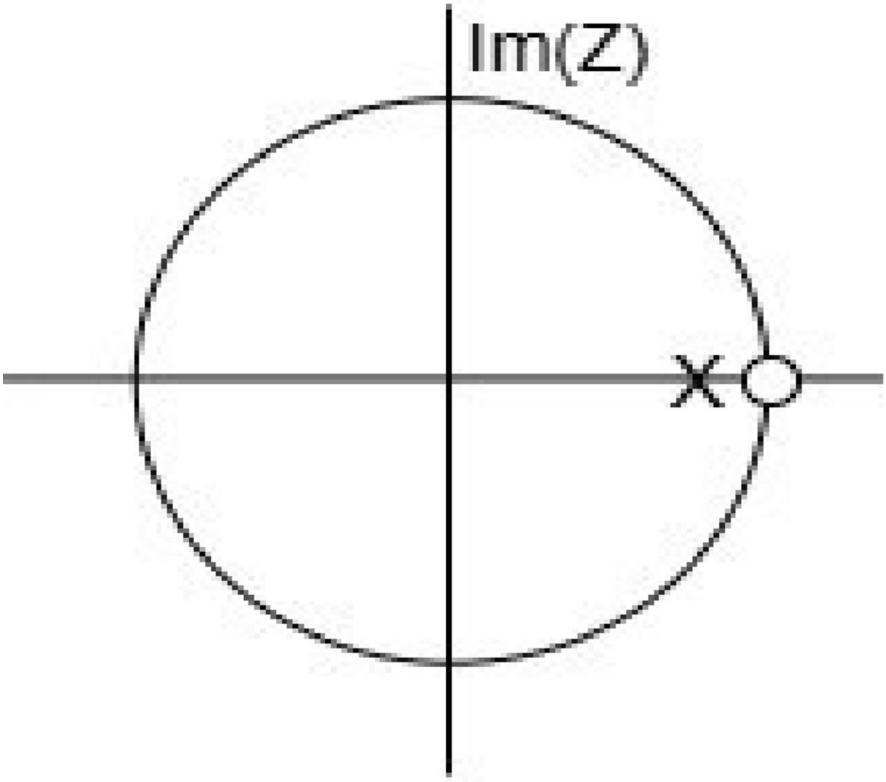


Fig. 2 Pole-zero placement for filter-1

3.1.2 Filter-2 to Allow Circulatory and Respiratory Systems Sounds and Suppress Background

Purpose: With the elimination of motion artifacts, the second filter is then tasked with isolating the frequency band in which circulatory (heart) and respiratory (lung) sounds exist, while at the same time eliminating other types of background noise. Noise may be ambient room noise, ambient environmental chatter, or even internal electronic noise generated by the device itself that exists beyond the desired physiological range.

Design Rationale: This filter is a low-pass filter. Its lower cut-off frequency is set to pass heart sounds (which contain major components at lower frequencies than lung sounds) and all pertinent lung sounds. Its upper cut-off frequency is set to exclude higher-frequency noise that is not diagnostically important for typical auscultation, e.g., hiss or high-frequency environmental noise.

Operation: Through the production of a "window" in the frequency spectrum, this filter maintains the main diagnostic sounds, but removes a wider frequency range of non-physiological background noise. This is a critical step to concentrate the sig-

nal on the most diagnostically important information and prepare it for the ultimate, more specialized extraction of heart sounds.

The respiratory and circulatory systems produce low-frequency (<500 Hz) noise. Most of the hospital background noise, especially from conversation, occurs at frequencies greater than 500 Hz. A second order IIR filter to cut this background noise is formed by choosing two poles and two zeros. Figure 3 shows where the poles and zeros are positioned on the pole-zero plot for this filter.

Poles are placed at 250 Hz and −250 Hz since the passband spans the frequency range of 0–500 Hz. For 250 Hz, the discrete-time frequency is $2\pi\,(250/8000)=\pi/16$. Likewise, the discrete-time frequency for −250 Hz is $2\pi\,(-250/8000)=-\pi/16$. As a result, the poles are placed as follows:

$$\mathrm{P}_0 = 0.9\mathrm{e}^{\mathrm{j}\omega 250} \quad \text{and} \quad \mathrm{P}_1 = 0.9\mathrm{e}^{-\mathrm{j}\omega 250} \tag{3}$$

The frequency range of the stopband would be 550–4000 Hz. The zeros may be positioned at an angle on the unit circle within this range and it's negative. We select 2000 Hz and −2000 Hz with zeros at $\pi/2$ and $-\pi/2$. The transfer function that results from a typical filter can be expressed as follows:

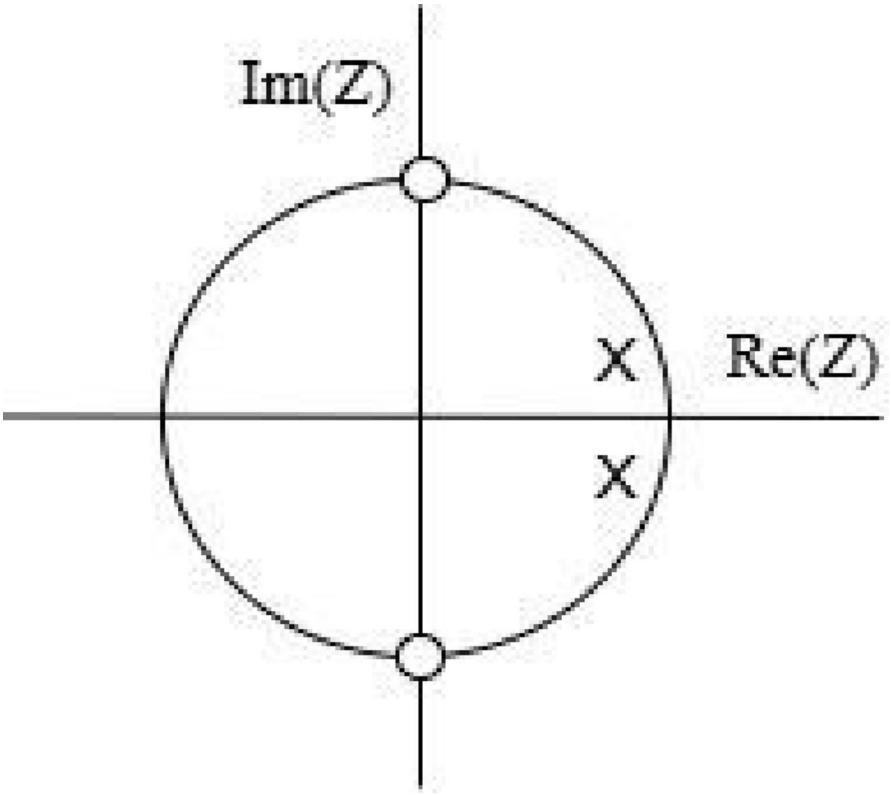


Fig. 3 Pole-zero placement for filter-2

$$H_2(z) = \frac{y_2(z)}{y_1(z)} = \frac{1 + z^{-2}}{1 - \frac{7}{4}z^{-1} + \frac{49}{64}z^{-2}} \tag{4}$$

and the desired output in the time domain is expressed as,

$$y(n) = x(n) - x(n-2) + \frac{7}{4}y(n-1) - \frac{49}{64}y(n-2) \tag{5}$$

3.1.3 Filter-3 to Allow Heart Beats

Purpose: The last filter in the cascade is designed in particular to pick out the discrete elements of heart sounds (S1, S2, and possibly S3, S4 murmurs) from the previously cleaned and band-limited signal by the previous stages. This accurate extraction is crucial for precise PCG analysis as well as for detecting cardiac defects.

Design Rationale: Heart sounds usually take up a particular range of frequencies, which happens to overlap with part of the lung sounds. This filter is thus designed to be another band-pass filter, but with a tighter passband, carefully placed around the usual frequencies of heart sounds. Its design balances carefully between the requirement of strong heart sound detection and wanting to eliminate any remaining lung sounds or noise which may still occur within its frequency range.

Operation: This filter is a refinement stage, zeroing in on the cardiac signals. By isolating the heart sound frequencies, it makes the typical "lub-dub" more distinct and refined, allowing for automated analysis or assisting clinicians in detecting faint heart murmurs and other abnormalities that would otherwise be obscured by noisy lung sounds or lingering background noise.

Besides digestive and respiratory sounds, the digital stethoscope also comes with a heartbeats listening mode. A second-order IIR is implemented to filter out heart sounds between 120 and 500 Hz using two zeros and two poles. The bandpass filter's bandwidth is 120–500 Hz. Poles have to be conjugate symmetric because of the occurrence of both positive and negative frequency components. The positions of the poles and zeros on the pole-zero diagram for this filter are indicated in Fig. 4.

Poles are positioned at 310 Hz and −310 Hz, or (120+500) Hz/2. For 310 Hz (−310 Hz), the discrete-time frequency is 31π/400 (−31π/400). Additionally, zeros are added by setting z=+1 and z=−1, which eliminates 0 and rad/sample. Hence, the poles are located as follows:

$$P_0 = 0.9e^{j\omega 310} \text{ and } P_1 = 0.9e^{-j\omega 310} \tag{6}$$

The desired filter's transfer function becomes,

$$H_3(z) = \frac{y_2(z)}{y_1(z)} = \frac{1 - z^{-2}}{1 - \frac{7}{4}z^{-1} + \frac{49}{64}z^{-2}} \tag{7}$$

and the time-domain output is expressed as,

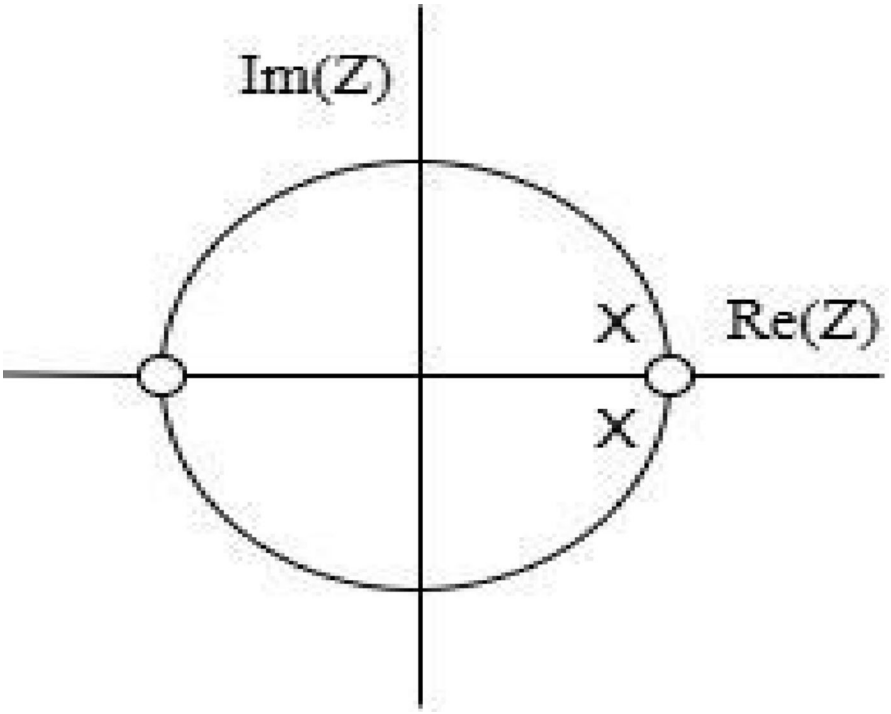


Fig. 4 Pole-zero placement for filter-3

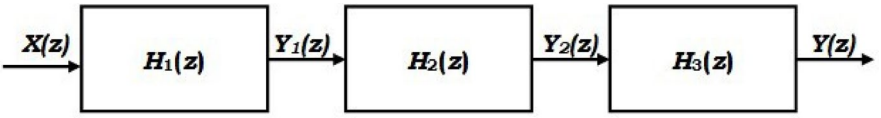


Fig. 5 Cascaded filter stages for the stethoscope

$$y(n) = x(n) - x(n-2) + \frac{7}{4}y(n-1) - \frac{49}{64}y(n-2) \tag{8}$$

3.1.4 Cascaded Digital Filters

The whole system has three filters. The input to the first filter is X(Z), its transfer function is $H_1(z)$, and the output response is $Y_1(Z)$. The output $Y_1(Z)$ is employed as the input to the second filter, where its transfer function is $H_2(z)$, and the output response is $Y_2(Z)$. Likewise, the output $Y_2(Z)$ is the input to the third filter whose transfer function is $H_3(z)$ and whose final output response is Y(Z) of the overall cascaded filter unit (see Fig. 5).

3.2 Numerical Representation and Arithmetic

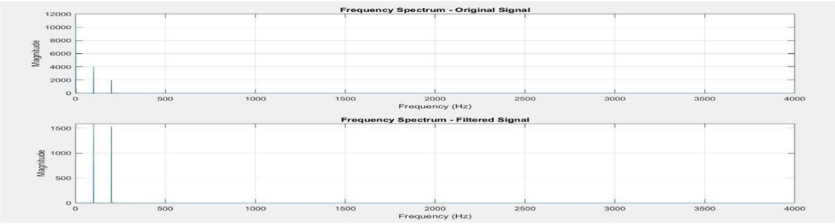
The cascaded digital filter of this paper uses a fixed-point numerical format based on two's complement integers for all internal operations. More particularly, the bits16 subtype provides a signed integer range over 16 bits. This was the choice taken following deliberate consideration of a number of factors of relevance to implementation in FPGA and the particular goals of this design. One important alternative to numeric representation in digital signal processing, especially for multiplication and division operations, is the Logarithmic number system (LNS). LNS has the unique benefit of reducing multiplication and division to addition and subtraction, respectively, thus avoiding rounding errors inherently present with fixed-point or floating-point multiplication and division [1]. In addition, fast FPGA implementations of algorithms that make use of LNS have already been shown [5, 23].

Our decision to implement fixed-point arithmetic in this design was based on practical considerations such as ease of implementation, guaranteed area and timing behavior, and extent of study. Fixed-point arithmetic takes direct advantage of optimized adder, subtractor, and shifter blocks available in current FPGAs, while logarithmic number system (LNS) needs additional computing resources for logarithm and antilogarithm conversion, which increases complexity. Fixed-point operations provide deterministic resource use and propagation delay, essential for achieving hard performance requirements, while LNS conversions introduce slice usage and increase clock-to-output delay. Our objective was to show the viability and efficiency of a purely integer-based digital filter cascade in fulfilling throughput, dynamic range, and total harmonic distortion requirements under constrained hardware resources.

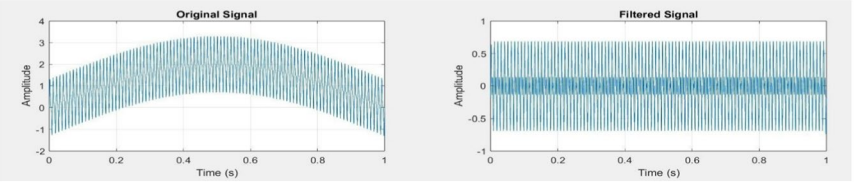
4 Software Simulations

A cost-effective and easy-to-use design exploration, parameter adjustment, and functional verification solution can be achieved by simulating the proposed work in a software environment before proceeding to hardware implementation. MATLAB provides robust environments for signal processing, as well as tools for digital filter modeling, analysis, and visualization. The simulation results include frequency spectra, time-domain plots, frequency responses, and pole-zero diagrams for the respective filters, which are shown in Figs. 6, 7 and 8.

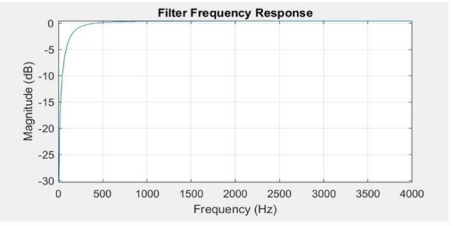
Before implementing resource-intensive hardware, this first simulation step assisted us in identifying potential design defects, assessing performance metrics under various scenarios, and fine-tuning the system architecture. A high-performance, real-time digital signal processing system for the digital stethoscope was made possible thanks in large part to the FPGA implementation. The design and execution of a pipelined architecture that greatly increased operating speed and decreased latency was made possible by utilizing the intrinsic benefits of FPGAs, such as their capacity for parallel processing, the potential for low power consumption, and architectural flexibility.



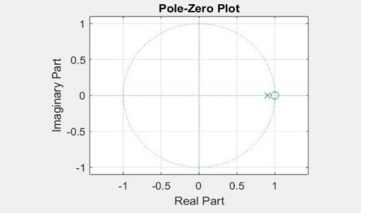
a



b



c



d

Fig. 6 **a** Frequency spectra of the original and filtered signals of filter1. **b** Time domain plots of the original and filtered signals for Filter-1. **c** Frequency response of Filter-1. **d** Pole-zero diagram of Filter-1

4.1 MATLAB Implementations

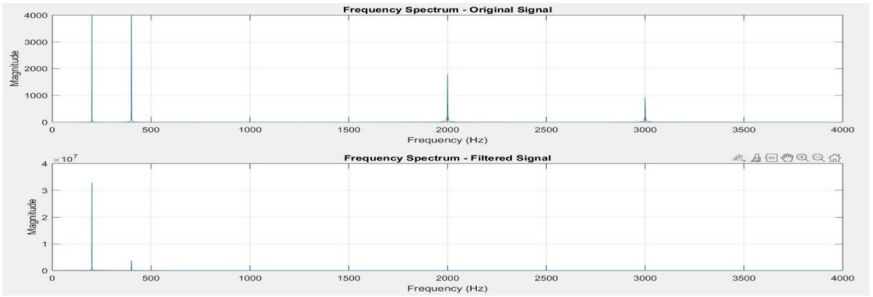
A MATLAB simulation platform was used in an effort to simulate the filter's performance. The parameters used in simulation were as follows:

4.1.1 Filter 1: Filter to Reject the Patient's Body Movement

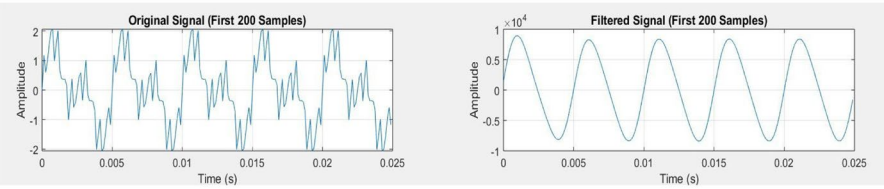
- Sampling frequency (fs): 8000 Hz
- Pole location (z): 0.9
- Zero location (z): 1

the above parameter were utilized to calculate the following filter coefficients:

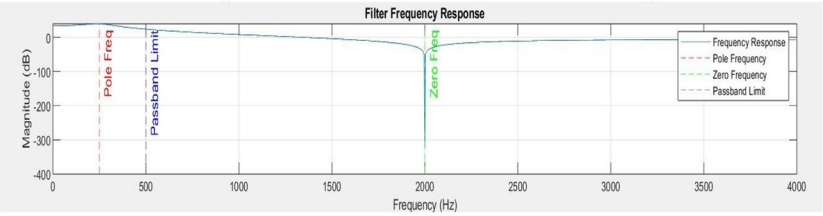
- Numerator coefficients (b): [1, −1]
- Denominator coefficients (a): [1, −0.9]



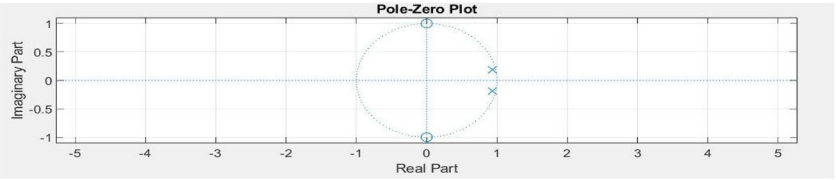
a



b



c



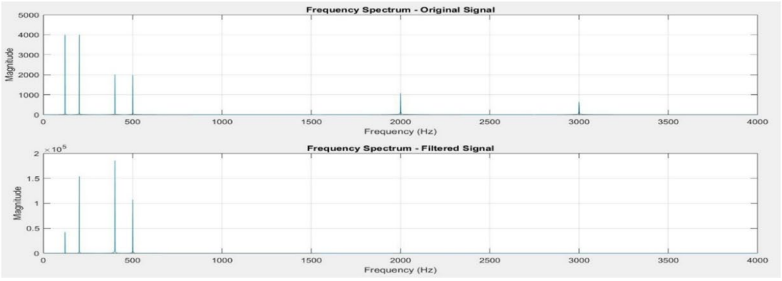
d

Fig. 7 **a** Frequency spectra of the original and filtered signals of filter2. **b** Time domain plots of the original and filtered signals for Filter-2. **c** Frequency response of Filter-2. **d** Pole-zero diagram of Filter-2

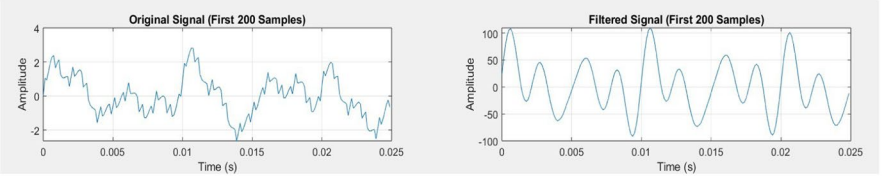
A test signal was produced to mimic actual data, and they were:

- Motion artifact at low frequency (0.5 Hz)
- Higher frequency components at 100 Hz and 200 Hz

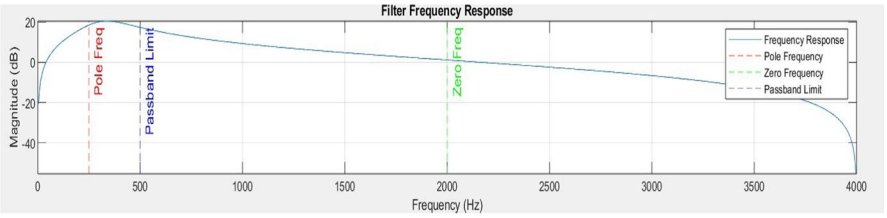
For zero-phase filtering, the filter was used on the test signal using the *filtfilt* function.



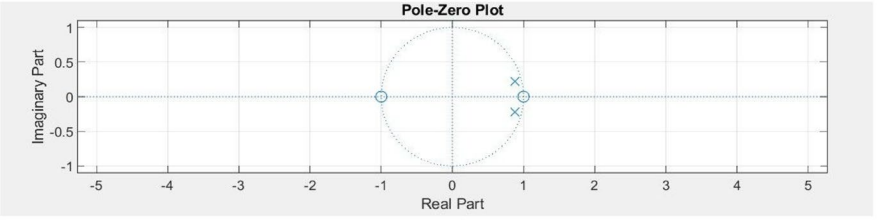
a



b



c



d

Fig. 8 **a** Frequency spectra of the original and filtered signals of filter 3. **b** Time domain plots of the original and filtered signals for Filter-3. **c** Frequency response of Filter-3. **d** Pole-zero diagram of Filter-3

4.1.2 Filter 2: Filter to Permit Circulatory and Respiratory Systems Sounds and Minimize Background

To attenuate higher frequency components and pass frequencies below 500 Hz, a second-order IIR filter was employed and designed. The parameters of the filter were chosen as follows:

- Filter type: Second-order IIR
- Pole positions: ±250 Hz (just on unit circle)
- Zero locations: ±2000 Hz (on unit circle)
- Passband: 0–500 Hz
- Stopband: 550–4000 Hz

For providing real-valued output, the filter implementation used conjugate pairs of zeros and poles. To provide stability, poles were placed slightly inside the unit circle (radius = 0.95). The filter is intended to attenuate higher frequency content above 550 Hz, background noise, while passing through frequencies below 500 Hz, respiratory and circulation sounds, very effectively. Attenuation is also good at about 2000 Hz.

4.1.3 Filter 3: Filter to Allow Heart Beats

A second-order IIR filter was utilized to select circulatory heart sounds between 120 and 500 Hz. The poles were placed at conjugate symmetric locations to act as a band-pass filter with a passband 120–500 Hz. The zeros were set at Z0 = 1 and Z1 = −1 to eliminate the DC component (0 rad/sample) and Nyquist frequency component (π rad/sample).

The filter developed has the following characteristics:

- Passband: 120–500 Hz
- Stopband: 0–120 Hz and 500–4000 Hz
- Pole radius: 0.9

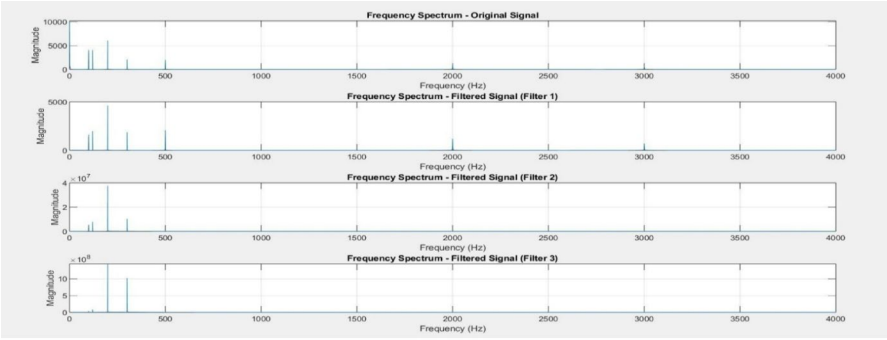
4.1.4 Cascaded Filter Design for Heart Sound Extraction and Noise Reduction

To obtain heart sounds efficiently and reduce noise levels in a digital stethoscope system, a cascaded filter structure was established. Three stages constituted the cascaded filter:

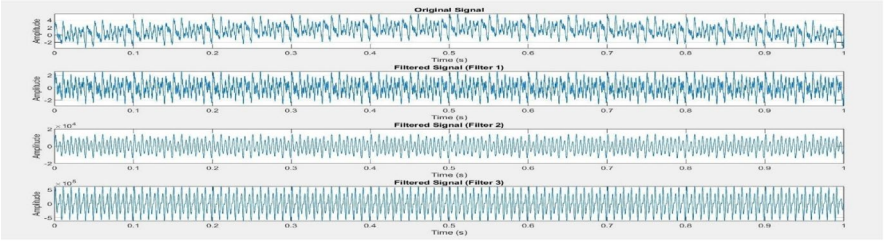
1. DC notch IIR filter (Filter 1)
2. Low-pass IIR filter (Filter 2)
3. Band-pass IIR filter (Filter 3)

To accomplish zero-phase filtering, the *filtfilt* function was used to implement the cascaded filter.

The output from one stage of the filter was sent to the next stage. An experimental signal, which included background noise, motion artifacts, and heart sounds, was used to evaluate the performance of the cascaded filter. The results demonstrated that the cascaded filter effectively removed the noise components while preserving the heart sounds. The original and filtered signals were graphically represented in both the time domain and the frequency spectrum to illustrate the effectiveness of the cascaded filter as shown in Fig. 9. Overall, the results indicated that the cascaded filter successfully eliminated the noise while maintaining the integrity of the heart sounds.



a



b

Fig. 9 **a**Time domain plots of the original and filtered signals for cascaded filters. **b** Frequency spectra of the original and filtered signals

The MATLAB output for three digital filters designed in this work is presented in Table 1. Each row includes the filter number, input and output signals, as well as the coefficients for the numerator and denominator. The input and output signals are discrete-time sequences, and the coefficients define the filter's transfer function.

4.2 FPGA Implementations Using VHDL Programming

The use of field-programmable gate arrays (FPGAs) was essential to this study, enabling the development of a high-speed, real-time digital signal processing system for digital stethoscopes. We designed and implemented both non-pipelined and pipelined architectures, with the pipelined architecture demonstrating improved operational speed and reduced latency. This enhancement was made possible by leveraging the advantages of FPGAs, such as real-time processing, high-speed performance and flexibility. The deployment of FPGAs will lead to improved patient outcomes and greater efficiency in healthcare services. It facilitated the automation of signal processing tasks, improved diagnostic accuracy, and enhanced patient care.

The design centers around a "cascaded filter unit", which consists of a series of filters where the output of one stage becomes the input for the next stage. The filter input, denoted as x_in, is given by the sequence of samples: {1000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}. This impulse-like input is commonly used to define the impulse response of a filter in various applications. The simulated output of the cascaded filter unit

Table 1 Mat lab output for filter transfer functions

Filter no.	Filter co-efficient	Inputs to filter	Outputs from filter
1	Numerator: 1, −1 Denominator: 1, −15/16	{1000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}	{1000, −62.5, −58.6, −54.9, −51.5, −48.3, −45.3, −42.4, −39.8, −37.3, −35}
2	Numerator: 1, 0, 1 Denominator: 1, −7/4, 49/64	{1000, −62.5, −58.6, −54.9, −51.5, −48.3, −45.3, −42.4, −39.8, −37.3, −35}	{1000, 1687.5, 3128.9, 4066.2, 4610.1, 4863.5, 4706.1, 4426.9, 4064.4, 3648.5}
3	Numerator: 1, 0, −1 Denominator: 1, −7/4, 49/64	{1000, 1687.5, 3128.9, 4066.2, 4610.1, 4863.5, 4706.1, 4426.9, 4064.4, 3648.5}	1000, 3438, 3128.9, 4066.2, 4610.1, 4851.3, 4863.5, 4706.1, 4426.9, 4064.4, 3648.5

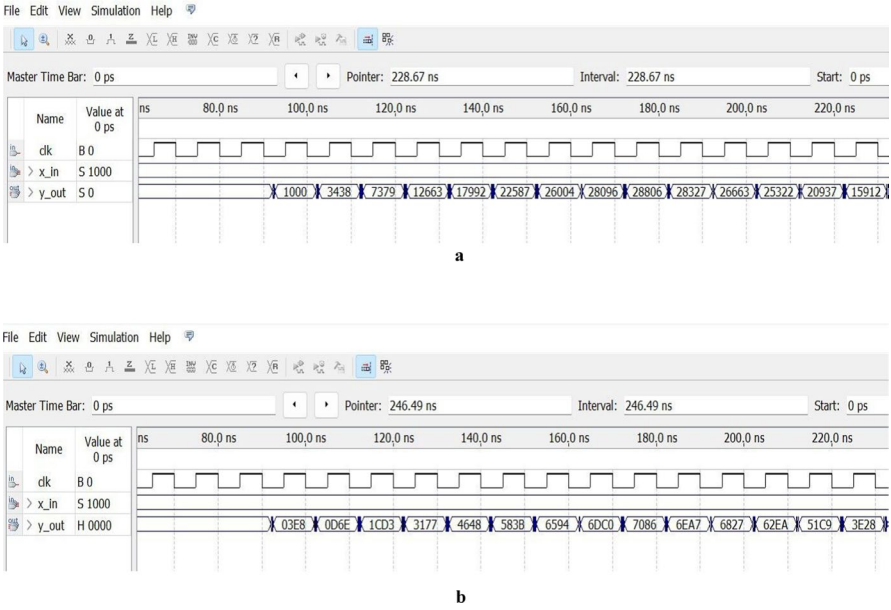


Fig. 10 **a** VHDL programming output using Quartus II software of the cascaded filter unit for the input samples {1000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0} in signed decimal format (conventional-pipeline). **b** VHDL programming output using Quartus II software of the Cascaded filter unit for the input samples {1000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0} in hexadecimal format (conventional-pipeline)

using Quartus II tool is illustrated in Fig. 10a, where the response of the filter is represented as y_out.

Using VHDL programming on the Cyclone II FPGA Board, we successfully created the block diagram, conducted simulations, and implemented the hardware with the in-system memory content editor for various modules, as detailed in Table 2. For

Table 2 Details of the implemented modules

Module no.	Conventional non pipelined	Conventional pipelined	Direct form-I non pipelined	Direct form-I pipelined
1	Filter 1	Filter 1	Filter 1	Filter 1
2	Filter 2	Filter 2	Filter 2	Filter 2
3	Filter3	Filter3	Filter3	Filter3
4	Cascaded filter	Cascaded filter	Cascaded filter	Cascaded filter

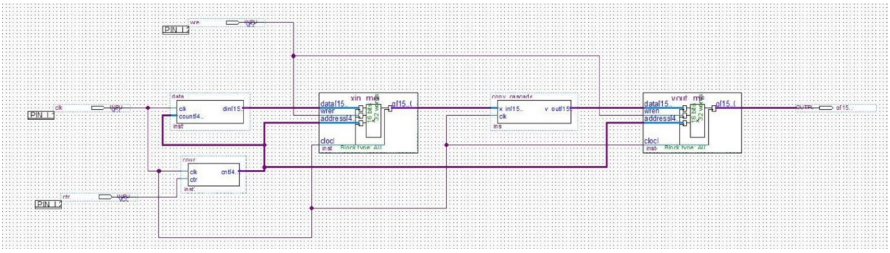


Fig. 11 Block diagram implementation of pipelined conventional form cascaded filter in Quartus II software

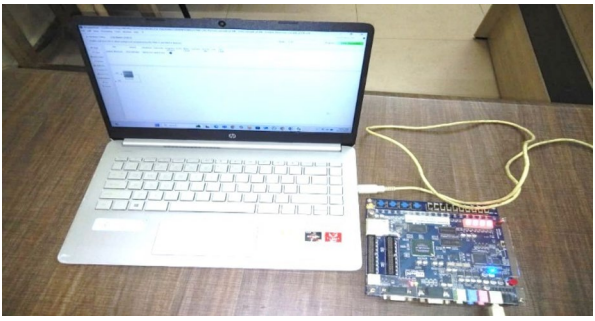


Fig. 12 Typical setup for using Quartus software with cyclone II FPGA Board for hardware implementation

demonstration purposes, we will focus on the implementation results of the conventional pipelined cascaded filter, as illustrated in Figs. 10, 11, 12 and 13. The procedures for the other modules are similar.

Using Quartus II version 13.0 service pack 1 software, the proposed cascaded digital filter module was synthesized on an FPGA platform, specifically the Altera/Intel Cyclone II 2C20F484C7 device, through programming in VHDL. Two different filter topologies—Classic and Direct Form I were implemented to evaluate their performance and usability. Additionally, both pipelined and non-pipelined design strategies were employed for these topologies. Pipelining techniques were used to enhance the operating speeds of the filters by introducing extra registers into the data stream, allowing for the simultaneous processing of different data samples.

We tested the response of all three digital filters individually, as well as the cascaded filter, using Quartus II 13.0 SP1 software. This involved implementing both conventional and direct form I structures in both non-pipelined and pipelined ver-

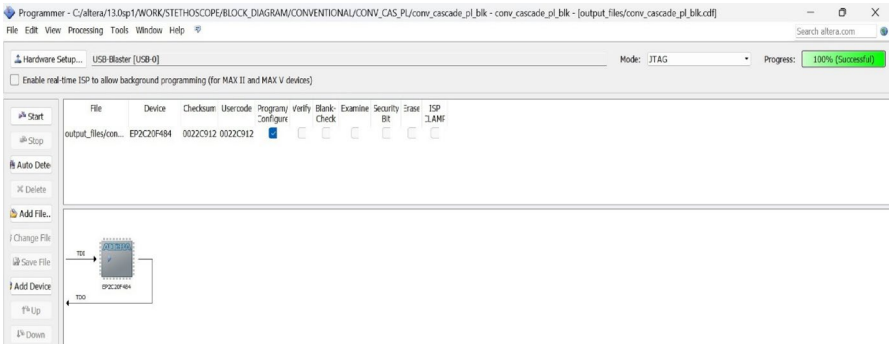


Fig. 13 Quartus II programmer interface

sions. In every case, the results we obtained closely matched those from MATLAB for each scenario. Figure 10a and b display the output of one version of the cascaded filter, specifically the conventional pipelined version.

5 Hardware Implementation

To implement the hardware design of the stethoscope filters, a block schematic of the design was created using Quartus II 13.0 SP1 software, which incorporates the In-System Memory Content Editor feature. The block diagram (Fig. 11) comprises block symbols for the cascaded stethoscope filter unit (conventional pipelined version), a 5-bit counter for address generation and data incrementing, an internal read/write memory block serving as input memory, and a read-only memory block acting as output memory.

5.1 Hardware Setup

The Quartus II 13.0 SP1 software toolset provides the Mega Wizard Plug-In Manager, a simple but effective tool that enables FPGA designers to easily design complex intellectual property (IP) cores and custom hardware blocks without requiring low-level hardware description language (HDL) programming. The tool enables the generation of various types of on-chip memory including RAM, ROM, FIFO buffers, and dual-port RAM with parameterizable sizes, data widths, address widths, and access modes. Input and output memory blocks were generated and added to the block diagram in our design using the Mega Wizard Plug-In Manager capability. The in-system memory content editor capability enables runtime monitoring and programming of the memory content via the output pins to facilitate a reconfigurable and flexible hardware platform for the stethoscope filter.

Figure 12 illustrates a typical setup for programming the Cyclone II FPGA Board using Quartus software for hardware implementation and Fig. 13 illustrates the FPGA programming process using the design from the output file with a USB-Blaster cable in JTAG programming mode. The message "*100% Successful*" indicates that the programming completed without any errors

5.2 In-System Memory Content Editor

To effectively examine and manage the behavior of the design without intruding upon it, we utilized the Quartus II In-System Memory Content Editor to view and modify the contents of memory blocks inside the FPGA in real time. This was made possible through the JTAG interface. Further developments were undertaken to enable in-system editing of memory content:

1. *Naming Memory Instances as Modifiable*: Memory instances created using the Mega Wizard Plug-In Manager were designated as "in-system modifiable."
2. *Compilation*: A complete compilation of the Quartus II project was conducted to include the necessary JTAG framework for accessing these memory blocks.
3. *Opening the In-System Memory Content Editor*: After launching the In-System Memory Content Editor, the Instance Manager was used to select the desired memory instance.
4. *Using the In-System Memory Content Editor*: The real-time editing and analysis were facilitated by displaying the memory contents within a Hex Editor pane.

5.2.1 Benefits of In-System Memory Content Editor

The direct access provided by the Quartus In-System Memory Content Editor to the internal memory of the FPGA significantly enhanced the hardware verification process. It enabled immediate feedback on the functionality of the implemented filter and facilitated effective debugging. At the time of FPGA programming, memory contents were preloaded using Memory Initialization Files (.mif). These files specify the initial values for each memory address, as well as the radix for both addresses and data, along with the depth and width of the memory. The success of the design was demonstrated through the efficient verification and validation of the hardware implementation of the proposed cascaded digital filter unit, utilizing the In-System Memory Content Editor (see Fig. 14).

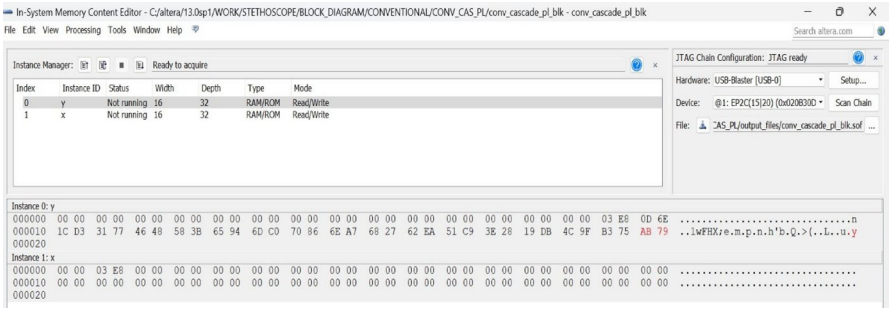


Fig. 14 Hardware Implementation output using cyclone II FPGA 2C20F484C7 device and in-system memory content editor of the cascaded filter unit for the input samples {1000, 0, 0, 0, 0, 0, 0, 0, , 0, 0, 0}

6 Results and Discussion

The digital stethoscope filters were implemented using two different structures: Conventional and Direct Form I, both with and without pipelining versions. All designs were synthesized using Quartus II software version 13.0 sp1 on an Altera Cyclone II FPGA (2C20F484C7) device. The VHDL programming language was used for all models. A comparative study of the simulation results is presented in Fig. 15a–c, evaluating the designs based on three performance metrics.

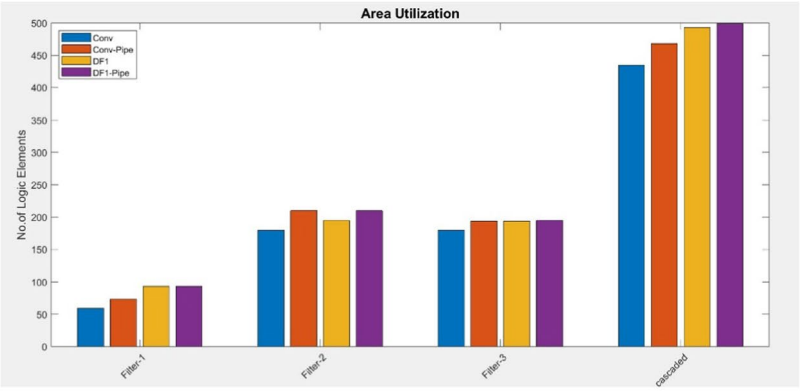
- Area utilization (No. of logic elements and registers)
- Performance (clock speed)
- Power consumption

The results clearly demonstrate the trade-offs between pipelined and non-pipelined digital filter implementations. For all filter structures (Conventional and Direct Form-1) and orders (Filter-1, Filter-2, Filter-3, and Cascaded Filters), pipelining consistently achieved substantial improvements in operating speed, generally exceeding 80%. However, this increased throughput typically required more hardware resources, as indicated by the greater number of Logic Elements (LEs) used in the pipelined designs. The impact on power consumption was more variable; while most pipelined implementations showed a modest to moderate increase in power usage, some Direct Form-1 structures demonstrated little or even slight power savings. These findings confirm that pipelining is effective for high-speed applications that require faster processing rates, although this comes with trade-offs related to area overhead and potential power implications depending on the specific filter architecture.

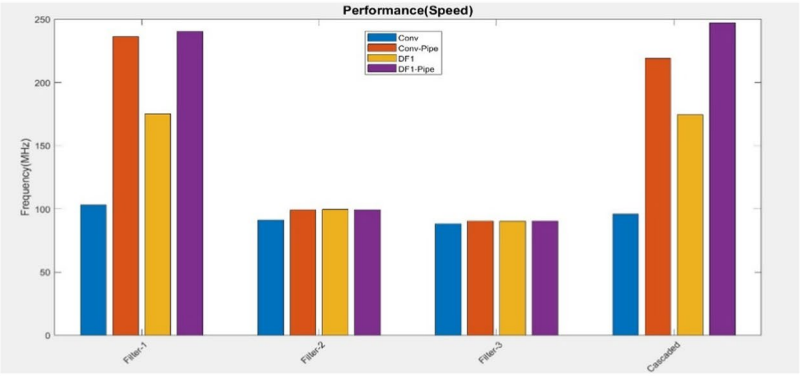
Though digital filters are well known to be significant in signal processing, and FPGAs are well-known for their benefits in high-performance computing, the innovation of our work is not simply in using these proven technologies, but rather in their specific, optimized combination to address one particular, compelling issue in digital auscultation. Our achievements extend the state of the art of digital stethoscope capability by addressing central deficiencies in a multi-faceted manner.

The comparison between the developed cascaded digital filters and other conventional biomedical signal filtering approaches is given in Table 3. This table emphasizes the different noise removal approaches applied to different biomedical signals such as PCG signals, ECG signals, and heart sound signals. The suggested cascaded digital filters for digital stethoscopes are unique because they utilize pipelined and non-pipelined architectures of cascaded IIR digital filters, using both the MATLAB and FPGA platforms, to efficiently eliminate motion artifacts, background noise, and non-heart sounds.

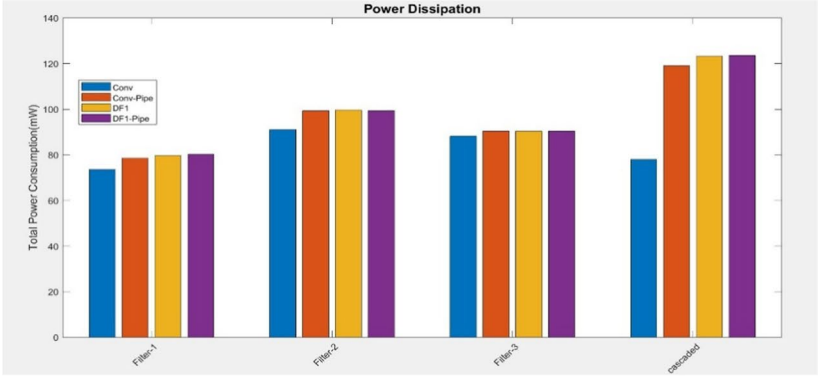
The suggested cascaded digital filters have better performance regarding noise elimination and signal quality improvement compared to other methods. For example, the denoising of PCG signals with wavelet transform and time delay neural network [4] and the application of spectral analysis in fake crackle noise research for electronic stethoscopes [5] are hopeful but restricted to certain noises. On the other hand, the cascaded filters discussed here provide a more general solution to auscultation improvement. The cascaded digital filters proposed for digital stethoscopes are



a



b



c

Fig. 15 **a** Area utilization report. **b** Performance report. **c** Power consumption report

Table 3 Comparative analysis of proposed cascaded digital filters and existing biomedical signal filtering techniques

Refs.	Research focus	Core technique	Implementation platform	Kind of noise removal
Proposed	Cascaded digital filters for digital stethoscopes to enhance auscultation	Pipelined and non-pipelined architectures of cascaded IIR digital filters	MATLAB (simulation), FPGA (hardware implementation)	Motion artifacts, Background noise and non-heart sounds
[9]	Adaptive denoising of PCG signals for mobile health monitoring	Wavelet transform + time delay neural network	MATLAB	White noise, pink noise, physiological and environmental disturbances
[14]	Investigation of fake crackle noise in electronic stethoscopes	Spectral analysis using high-pass and low-pass filtering; manual frequency band deletion	3M Littmann 3200 electronic stethoscope; Audacity software; SPSS for stats	Manual spectral filtering
[2]	Removal of baseline wandering noise from ECG signals	High-speed IIR filters with pipelining, look-ahead, and clustered look-ahead	FPGA	Baseline wandering (low-frequency DC bias)
[4]	Detection and removal of artifacts in phonocardiogram signals to improve cardiac auscultation accuracy	Fusion of tunable Q-wavelet transform and signal second difference with median filter	MATLAB	Artifact removal
[18]	Enhancing cardiac auscultation accuracy using digital filters in electronic stethoscopes	Digital filtering using infinite impulse response algorithm	Arduino Mega 2560 microcontroller with Nextion TFT LCD display and Bluetooth headset	Environmental and signal noise via bandpass filtering and IIR digital filtering
[7]	Design and analysis of reduced-order IIR filters for ECG R-peak detection	Optimum reduced-order infinite impulse response filter using min–max optimization	MATLAB (Simulated on ECG datasets from PhysioNet)	Motion artifacts, baseline wander, power line interference, electromyography noise
[25]	Real-time heart rate signal detection using electronic stethoscope	Signal conditioning with pre-amplifier and 4th-order Butterworth filters	LabVIEW + Arduino + Bluetooth HC-05	High-pass and low-pass Butterworth filters to remove murmurs and ambient noise
[3]	Design and analysis of a low-cost electronic stethoscope	4th-order Butterworth bandpass filtering	Arduino Nano + Bluetooth HC-05 + PC	Bandpass filtering to remove ambient and irrelevant frequency noise
[16]	Mimicking acoustic stethoscope sound using electronic filtering	Band equalization via spectral analysis	MATLAB-based signal processing pipeline Respiratory sound simulator for data input	Spectral subtraction + bandpass filtering High-pass filtering

Table 3 (continued)

Refs.	Research focus	Core technique	Implementation platform	Kind of noise removal
[13]	ECG signal classification and QRS peak detection	Min–Max optimized reduced-order IIR filter	MATLAB + FPGA prototype	Baseline wander, power line interference, high-frequency noise
[6]	Enhancing ECG signal quality by detecting and filtering noise using supervised learning	Supervised learning-based noise detection using Adaboost classifier with temporal and spectral features	Python (Scipy library)	Baseline Wandering, Powerline Interference, Muscle Artifacts
[10]	Noise elimination from ambulatory ECG signals and R-R interval detection	discrete wavelet transform with soft and hard thresholding	Python (using Physionet databases: MIT-BIH, PTB)	Power line interference, baseline drift, electromyogram, motion artifacts, white Gaussian noise
[11]	Heart sound signal processing using electronic stethoscope	FFT, wavelet transform, hanning window	MATLAB & LabVIEW	50 Hz noise suppression
[12]	Denoising phonocardiogram signals for cardiac diagnosis	Sliding window LMS adaptive filter	MATLAB R2022A + Bluetooth Stethoscope	Stationary and nonstationary noise
[8]	Enhancing phonocardiogram signal quality for accurate cardiac diagnostics	Cascaded adaptive noise canceller optimized using greater cane rat algorithm	MATLAB	Uniform noise, Gaussian noise and pink noise
[19]	Measurement of heart and lung sounds during general anesthesia using a digital esophageal stethoscope	Adaptive noise cancellation using least mean square algorithm	Teensy 3.2 microcontroller + MATLAB App for signal display	Operating room noise

an important step in the field of biomedical signal processing, providing enhanced noise removal and signal enhancement. The comparison with other methods indicates the promise for real-time performance and the potential for integration into handheld digital stethoscope systems.

7 Conclusion

This research effectively developed, deployed, and rigorously verified a new FPGA-based cascaded digital filter architecture that dramatically improves auscultation in digital stethoscopes by efficiently suppressing unwanted noise and boosting vital body sounds. The newly conceived three-stage cascaded filter designed specifically to suppress motion artifacts and environmental noise was carefully verified through both MATLAB simulation offering critical functional validation and property checking and corresponding FPGA hardware deployment. One of the key findings of this research is the evidently superior performance attained with our pipelined FPGA implementation that has been optimized. Contrary to traditional or non-pipelined dig-

ital filter structures, our design maximizes the inherent parallelism of FPGAs to attain significant gains in processing rate and throughput. The result showed a dramatic increase in operating rate because of pipelining, most often more than 80%, across different forms and orders of the filter. Though greater throughput tends to have a cost in terms of greater hardware resource usage, as evidenced by greater Logic Element numbers in pipelined implementations, our method preserves impressive total resource utilization. In addition, the impact of pipelining on power consumption exhibited more nuanced behavior, with most designs experiencing a slight increase, albeit with some Direct Form-1 implementations recording negligible changes or even slight reductions. These findings strongly highlight the useful trade-off offered by pipelining in high-speed digital stethoscope applications, where processing speed is most critical. Our careful design, confirmed through detailed MATLAB simulations and demonstrated on an Altera/Intel FPGA 2C20F484C7 board with VHDL programming, underscores the real-world practicability and effectiveness of our methodology. Not only does this research introduce a measurable quality enhancement to digital auscultation but also establishes a new standard for optimizing performance, precision, and hardware footprint in real-time medical signal processing. Future research will concentrate on applying this filter unit into a fully functional digital stethoscope prototype for rigorous clinical trials, and investigating advanced adaptive filtering methods for further noise cancellation optimization in dynamic clinical settings.

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Data Availability The source code related to this research is available on GitHub https://github.com/dpfi0423011686-eng/STETHOSCOPE_FILTER.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

References

1. F. Albu, A recursive least square algorithm for active noise control based on the Gauss–Seidel method, in *Proceedings of the 15th International Conference on Control Systems and Computer Science (CSCS 15)* (2005), pp. 1–6. https://doi.org/10.1109/ICECS.2008.4674982
2. F. Albu, J. Kadlec, N. Coleman, A. Fagan, Pipelined implementations of the a priori error-feedback LSL algorithm using logarithmic arithmetic, in *2002 IEEE International Conference on Acoustics, Speech, and Signal Processing* (2002), pp. III-2681–III-2684. https://doi.org/10.1109/ICASSP.2002.5745200
3. M.R. Alla, C. Nayak, Enhancing PCG signal quality through cascaded adaptive noise cancelling with metaheuristic optimization. Circuits Syst. Signal Process. **44**, 7776–7815 (2025). https://doi.org/10.1007/s00034-025-03166-x

4. H. Amhia, A.K. Wadhwani, Stability and phase response analysis of optimum reduced-order IIR filter designs for ECG R-peak detection. J. Healthc. Eng. **2022**, 1–14 (2022). https://doi.org/10.1155/2022/9899899
5. D. Datta, H.S. Dutta, High performance IIR filter implementation on FPGA. J. Electr. Syst. Inf. Technol. **8**(2), 1–11 (2021). https://doi.org/10.1186/s43067-020-00025-4
6. D. Gradolewski, G. Redlarski, Wavelet-based denoising method for real phonocardiography signal recorded by mobile devices in noisy environment. Comput. Biol. Med. **52**, 146–153 (2014). https://doi.org/10.1016/j.compbiomed.2014.06.011
7. D. Gradolewski, G. Magenes, S. Johansson, W.J. Kulesza, A wavelet transform-based neural network denoising algorithm for mobile phonocardiography. Sensors **19**, 957 (2019). https://doi.org/10.3390/s19040957
8. Y. Kim, K.B. Kim, A.Y. Leem, K. Kim, S.H. Lee, Enhanced respiratory sound classification using deep learning and multi-channel auscultation. J. Clin. Med. MDPI **14**, 1–25 (2025). https://doi.org/10.3390/jcm14155437
9. A. Kishore Kumar, G. Saha, Improved computerized cardiac auscultation by discarding artifact contaminated PCG signal sub-sequence. Biomed. Signal Process. Control **41**, 48–62 (2018). https://doi.org/10.1016/j.bspc.2017.11.001
10. I. Kouretas, V. Paliouras, logarithmic number system for deep learning, in *2018 7th International Conference on Modern Circuits and Systems Technologies (MOCAST)* (2018), 1–4. https://doi.org/10.1109/MOCAST.2018.8376572
11. D. Kumar, P. Carvalho, M. Antunes, R.P. Paiva, J. Henriques, Noise detection during heart sound recording using periodicity signatures. Physiol. Meas. **32**(5), 599–618 (2011). https://doi.org/10.1088/0967-3334/32/5/008
12. S. Lohbare, S. Dixit, Elimination of noise from ambulatory ECG signal using DWT. Int. J. Eng. Trends Technol. **70**(5), 266–273 (2022). https://doi.org/10.14445/22315381/IJETT-V70I5P229
13. N. Manjula, N.P. Singh, P. Ashok Babu, An efficient designing of IIR filter for ECG signal classification using MATLAB, in *Engineering Proceedings (MDPI)* (2023), pp. 1–14. https://doi.org/10.3390/HMAM2-14154
14. N. Mohamed, H.-S. Kim, K.-M. Kang, M. Mohamed, S.-H. Kim, J.G. Kim, Heart and lung sound measurement using an esophageal stethoscope with adaptive noise cancellation. Sensors **21**, 6757 (2021). https://doi.org/10.3390/s21206757
15. R. Mohan Raj, V. Rajesh, S. Saravanan, M.S.P. Balaji, V. Elamaran, Baseline wandering noise removal using high-speed IIR filters with an FPGA implementation, in *ICMDCS 2021 (CCIS 1392)* (2021), pp. 55–65. https://doi.org/10.1007/978-981-16-5048-2_5
16. P. Oktivasari, F. Haryanto, A real-time heart rate signal detection using an electronic stethoscope with labview. J. Biomed. Phys. Eng. **10**(3), 375–382 (2020). https://doi.org/10.31661/jbpe.v0i0.1183
17. P. Oktivasari, R.A.F. Riandini, S.I. Malaon, Active filter analysis on designing electronic stethoscope. J. Elektron. Telekomun. JET **19**(2), 51–56 (2019). https://doi.org/10.14203/jet.v19.51-56
18. P. Oktivasari, H.S.R. Amy, H. Freddy, Suprijadi, Study and analysis of electronic stethoscope signal using MATLAB & LabVIEW as effective tools. J. Phys. Conf. Ser. **1505**, 1–10 (2020). https://doi.org/10.1088/1742-6596/1505/1/012054
19. V. Rennoll, I. McLane, D. Emmanouilidou, J. West, M. Elhilali, Electronic stethoscope filtering mimics the perceived sound characteristics of acoustic stethoscope. IEEE J. Biomed. Health Inform. **25**(5), 1542–1553 (2021). https://doi.org/10.1109/JBHI.2020.3020494
20. A.F. Rohman, M.R. Mak'ruf, T. Triwiyanto, L. Lamidi, P.H. Huynh, Analysis of the effectiveness of using digital filters in electronic stethoscopes. J. Electron. Electromed. Eng. Med. Inform. **4**(4), 229–234 (2022). https://doi.org/10.35882/jeeemi.v4i4.256
21. T.S. Roy, J.K. Roy, N. Mandal, Design of ear-contactless stethoscope and improvement in the performance of deep learning based on CNN to classify the heart sound. Med. Biol. Eng. Comput. **61**, 2417–2439 (2023). https://doi.org/10.1007/s11517-023-02827-w
22. V.M. Shervegar, A robust sliding window adaptive filtering technique for phonocardiogram signal denoising. Multimed. Tools Appl. **84**, 33935–33961 (2025). https://doi.org/10.1007/s11042-024-20564-6
23. V. Supraja, P.N. Rao, M.N.G. Prasad, Supervised learning-based noise detection to improve the performance of filter-based ECG signal denoising. SSRG Int. J. Electron. Commun. Eng. **10**(6), 35–51 (2023). https://doi.org/10.14445/23488549/IJECE-V10I6P105

24. P. Varady, Wavelet-based adaptive denoising of phonocardiographic records, in *2001 Conference Proceedings of the 23rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (IEMBS)*, vol. 2 (2001), pp. 1846–1849. https://doi.org/10.1109/IEMBS.2001.1020582
25. Y.-C. Wu, C.-C. Han, C.-S. Chang, F.-L. Chang, S.-F. Chen, T.-Y. Shieh, H.-M. Chen, J.-Y. Lin, Development of an electronic stethoscope and a classification algorithm for cardiopulmonary sounds. Sensors **22**, 4263 (2022). https://doi.org/10.3390/s22114263
26. P. Ye, Q. Li, W. Jian, S. Liu, L. Tan, W. Chen, D. Zhang, J. Zheng, Regularity and mechanism of fake crackle noise in an electronic stethoscope. Front. Physiol. **13**, 1079468 (2022). https://doi.org/10.3389/fphys.2022.1079468

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Authors and Affiliations

S. Devi Poonguzhali[1] · T. N. Prabakar[1] · V. Elamaran[1]

✉ T. N. Prabakar
prabakar@ece.sastra.edu

S. Devi Poonguzhali
dpfi0423011686@sastra.ac.in

V. Elamaran
elamaran@ece.sastra.edu

[1] Department of ECE, School of EEE, SASTRA Deemed University, Thanjavur, Tamil Nadu, India